

EquiPhase — Review Gate 0 Checklist

Audit of Antigravity's IMPLEMENTATION_PLAN.md before any data/code is written. Mark each item Pass or Reject. Any Reject, or any auto-reject trigger, sends the plan back for revision.

CRITICAL PRE-CHECK — data sufficiency. Confirm BEFORE anything else.

The entire H1 rests on the extrapolation regime and on quantitative condition-dependent data. If the public datasets do not actually contain enough quantitative saturation-concentration records with the condition coverage needed for a low-salt -> high-salt (or analogous) split, the benchmark is under-powered and the design must be revised now, not discovered in month 3. The plan must state the realistic counts per condition regime and confirm sufficiency, or propose a fallback task. This is a hard gate.

Check item	Result
A. Data and licensing	
Each dataset (LLPSDB v2, PhaSePred, DrLLPS, CD-CODE) named with its current release and licence verified; any source that cannot be legally used is flagged.	Pass / Reject
Access method is concrete (API/download/dump), not 'scrape somehow'. Field-level extraction mapping (which columns -> features / labels / conditions) is specified.	Pass / Reject
Honest data limitations stated: sizes, class imbalance, and CONDITION coverage for the extrapolation split (ties to the critical pre-check above).	Pass / Reject
B. Splits and leakage control	
Family split uses a concrete clustering method at a stated identity threshold (e.g., MMseqs2 at 30-40%), not a vague 'family' label.	Pass / Reject
Condition-regime split defined with actual numeric thresholds; temporal embargo plan present, or an explicit note that assay dates are unavailable.	Pass / Reject
Leakage audit method described (no near-duplicate sequences across splits; how checked). Locked-test guard: a single script, never imported by training code.	Pass / Reject
C. Pre-registration completeness	
PRE_REGISTRATION.json lists task definitions, exact metric formulas (or metrics.py refs), seed count, and the bootstrap-CI procedure.	Pass / Reject
The FULL three-part win condition is present and frozen: (i) extrapolation Spearman ≥ 0.05 with non-overlapping 95% CIs across ≥ 5 seeds; (ii) match-or-beat in-distribution; (iii) memory advantage quantified. (Antigravity's summary listed only (i).)	Pass / Reject
Null-result policy stated explicitly (report, do not retune). File is hashable + timestamped.	Pass / Reject
D. Baseline fidelity	
Each baseline has a concrete reimplementations or official-code plan; faithfulness check described (reproduce a published number where possible).	Pass / Reject

Check item	Result
Any baseline that cannot be faithfully reproduced is flagged, not silently approximated.	
E. DEQ specifics	
Implicit-layer library named and its availability/version verified (e.g., torchdeq). Plan to measure local Lipschitz L and log full convergence traces.	Pass / Reject
R3 anti-recycling check implemented in code: how identical residual-trace detection works and what HALT does.	Pass / Reject
F. Integrity tooling (R1-R7 in code)	
run_id + config hash + git SHA logging; metrics.py single-source-of-truth; DEVIATIONS.md mechanism; locked-test guard; an explicit 'report null' code path.	Pass / Reject
G. Compute and feasibility	
Fits a single RTX 4090: VRAM budget per model (esp. ESM-2 embeddings + DEQ). If not, a concrete plan (precompute embeddings / smaller ESM-2). Timeline realistic.	Pass / Reject
Risk register includes the data-coverage risk AND the 'DEQ does not converge / L >= 1' risk, each with a mitigation.	Pass / Reject
H. Open questions	
The plan ENDS with genuine open questions for the human. Zero open questions at this stage is itself a red flag (R7).	Pass / Reject

Auto-reject triggers (bounce immediately)

- Any concrete result, metric value, or convergence number already present — there must be none at Gate 0.
- Vague data access ('we will find the data') or an unverified/non-existent dataset or library.
- Win condition weakened, partial, or movable after freeze.
- No leakage-audit method, or training code that can import the locked-test.
- No open questions returned, or specs filled by assumption rather than by asking.
- Data-sufficiency for the extrapolation regime not demonstrated.

Sign-off

Gate 0 passes only when every category above is Pass and no auto-reject trigger fires. On pass, freeze PRE_REGISTRATION.json and proceed to Prompt 1 (data + splits). On any reject, return the specific items to Antigravity with required fixes and re-audit.

Auditor: _____ Date: _____ Decision: PASS / REVISE

— End of checklist —